

SixLight – Assembly Guide

This guide covers the assembly of the **SixLight** lamp structure only.

This is a short illustrated guide to the **SixLight** lamp structure only. Electrical wiring and the ceiling pulley system are not described here but referenced in the final images.

Part List

Part number	Quantity	Description
001	6	50x50x1000mm Alluminium profile 2mm thickness
002	6	3d printed frame joint
003	6	3d printed sphere
004	6	M8 x 165mm threaded bar
005	24	Hex socket head cap screw M5x0.80 x 65 Stainless Steel
006	6	Self-locking hex nut M8 Stainless Steel
007	24	Hex nut M5 Stainless Steel
008	6	M8 eyebolt
009	6	<i>Assembly of LED fixtures (cad reference)</i>
010	12	3d printed Caps
011	6	Industrial 120cm LED batten housing
012	6	30x30x640mm Alluminium profile 2mm thickness
013	12	Symmetrical half ball socket
014	6	Female threaded fixing ring

Part number	Quantity	Description
015	6	Male threaded fixing ring
016	24	M5 Stainless Steel washer
017	12	Hex nut M5 Stainless Steel
018	12	Hex socket head cap screw M5 x 50 Stainless Steel
019	1	30x30x1818mm Alluminium profile 2mm thickness
020	24	Aluminium rivet Ø3 mm × 10 mm
021	2	Aluminium rivet Ø5 mm × 10 mm

3D Printing Recommendations

All parts were printed using a Bambu Lab X1 Carbon with a 0.6 mm nozzle and basic PLA without any supports.

All components, except for the spherical joint, are printed with **100% infill** for maximum strength.

The sphere is printed with: - **5 perimeters** - **6 bottom layers** - **6 top layers** - **30% infill**

Since the joint relies on precise fit, it is recommended to **print one piece first** and test the assembly.

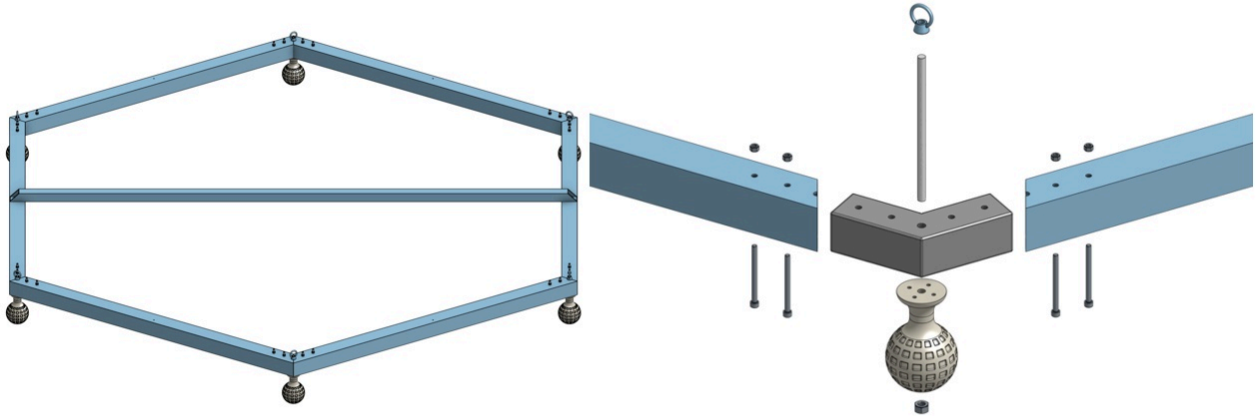
If the fit is too tight or too loose, adjust the slicer's **external dimension parameters** slightly until the desired tolerance is achieved.

!IMPORTANT - To ensure smooth threading and avoid jamming on the layer ridges of the printed part, components 013 (spherical jaw halves) and 015 (male threaded fixing ring) should be sanded with fine-grit sandpaper.

Frame Overview

The SixLight frame is composed of six 50x50 mm aluminum tubes arranged in a hexagon. At each corner, a 3D printed spherical housing is mounted.

Using carbon fiber tubes instead of aluminum can reduce the total frame weight by over 3 kg, while maintaining similar dimensions.



Parts used in the picture:

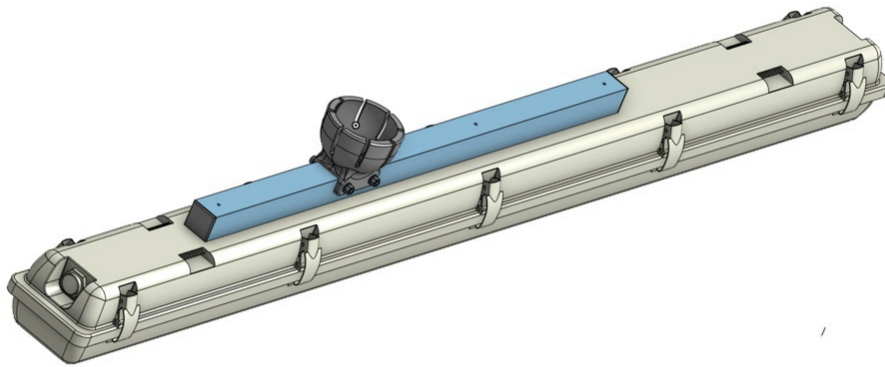
- **001** – 50x50x1000mm Aluminium profile 2mm thickness
- **002** – 3d printed frame joint
- **003** – 3d printed sphere
- **004** – M8 x 165mm threaded bar
- **005** – Hex socket head cap screw M5x0.80 x 65 Stainless Steel
- **006** – Self-locking hex nut M8 Stainless Steel
- **007** – Hex nut M5 Stainless Steel
- **008** – M8 eyebolt
- **019** - 30x30x1818mm Alluminium profile 2mm thickness

Apply threadlocker to the M8 eyebolt (part 008) and the M8 self-locking nut (part 006) during assembly to ensure long-term stability.

LED Fixture Mounting

Each LED batten is mounted on a reinforced aluminum tube (30x30 mm), riveted for strength.

At each end of the reinforcement bar, two spherical mount halves are fixed to form a housing for the ball joint.

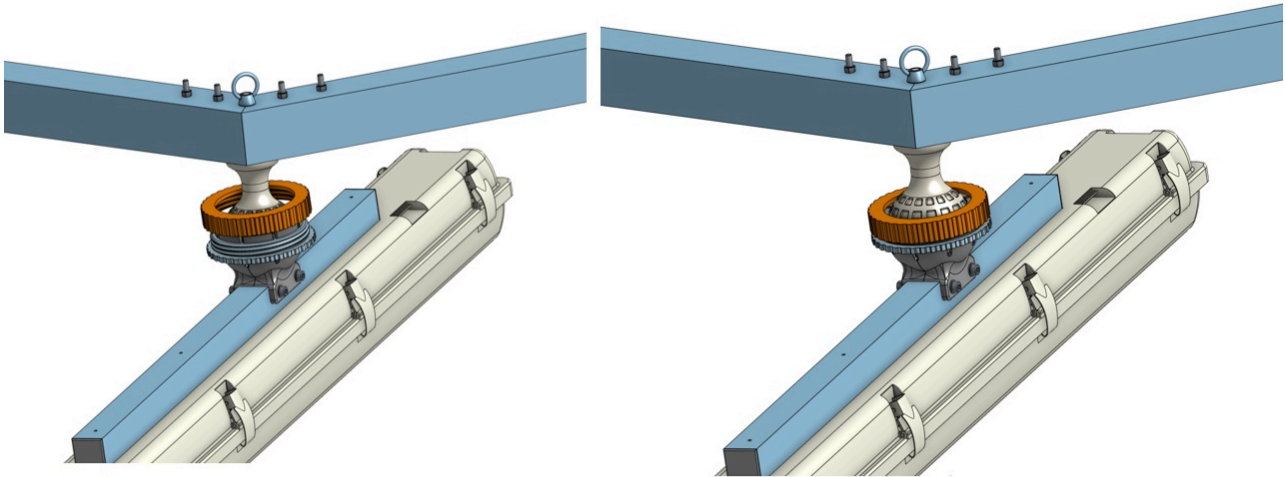


Parts used in the picture:

- **010** – 3d printed Caps
 - **011** – Industrial 120cm LED batten housing
 - **012** – 30x30x640mm Aluminium profile 2mm thickness
 - **013** – Symmetrical half ball socket
 - **016** – M5 Stainless Steel washer
 - **017** – Hex nut M5 Stainless Steel
 - **018** – Hex socket head cap screw M5 x 50 Stainless Steel
 - **020** – Aluminium rivet Ø3 mm × 10 mm
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Spherical Clamp Assembly

Each sphere fits into a set of two spherical jaws that cradle it securely. These are held together with a threaded clamping ring.



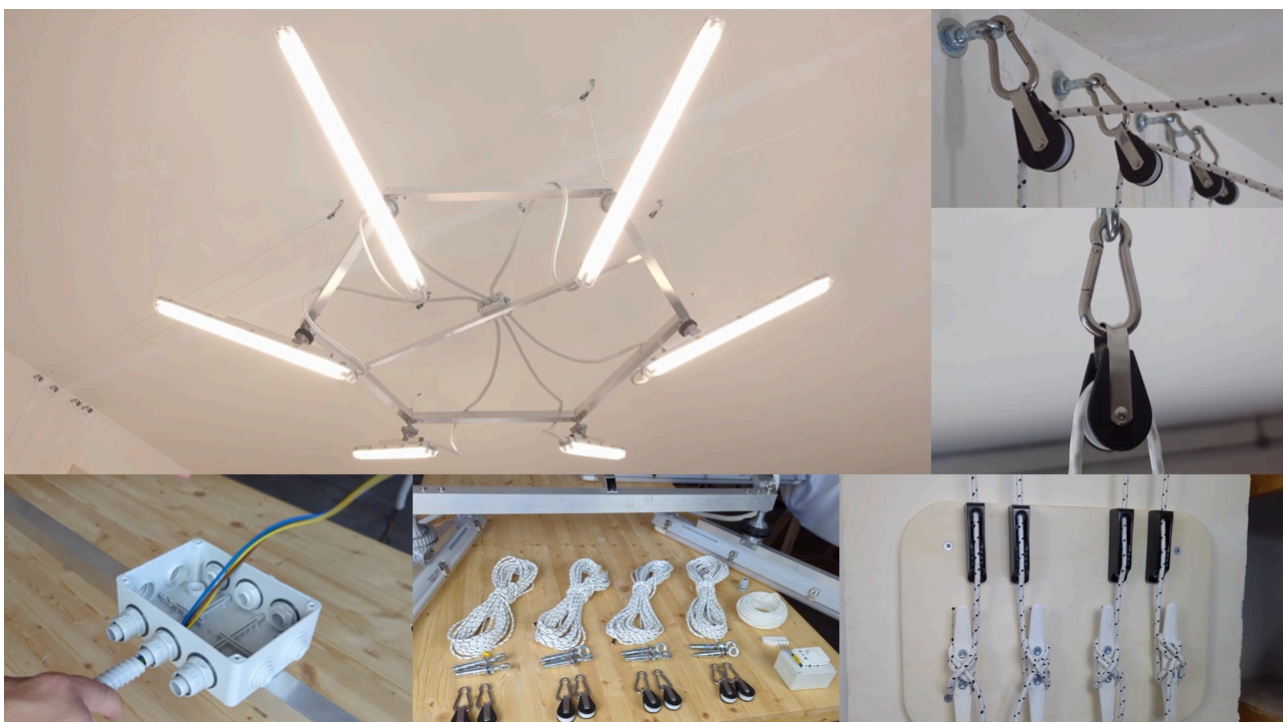
Parts used in the picture:

- **014** – Female threaded fixing ring
- **015** – Male threaded fixing ring

Once tightened, the lamp remains in the selected position.

The joint allows for a full 360° rotation and $\pm 30^\circ$ tilt.

Final Assembly & Installation Example



This version was fixed to the ceiling using a commercial pulley and cable system.
The electrical connection is made via standard 220V AC cabling (*see local regulations*).

Disclaimer

This guide is provided **as is**, without warranty.

Always ensure proper mechanical and electrical safety during construction and use.